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Compact and Ultra-Broadband 3 dB Power Splitter Based on Segmented Adiabatic Tapered Rib Waveguides

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Abstract: Optical 3 dB power splitters are fundamental building blocks for advanced silicon photonic integrated circuits, with applications ranging from high-speed modulators to optical phased arrays and programmable photonic processors. However, 3 dB power splitters are commonly hampered by trade-offs in device dimensions, operation bandwidth, and fabrication technology. In this paper, we present a compact and ultra-broadband 3 dB power splitter based on segmented adiabatic tapered rib waveguides, with a length of 23.4 μm . The simulated splitter achieved an output transmission efficiency exceeding 48% over a large wavelength of 400 nm (1200–1600 nm). The power splitter was successfully fabricated on a commercial platform and exhibited excellent splitting ratios within $50 \pm 3.8\%$ and insertion losses below 0.38 dB over the range of 1260–1360 nm and 1525–1600 nm. Additionally, a high-speed Mach–Zehnder modulator based on the power splitter was built, demonstrating 40 Gbps NRZ signal modulations across both O-band and C-band. The proposed splitter and modulator are promising elements for large-scale and broadband integrated photonic systems.

Keywords: power splitters; adiabatic tapers; taper profile optimization; silicon photonics



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1. Introduction

The rapid advancement of next-generation information technologies, such as artificial intelligence, big data, and cloud computing, has imposed significant challenges on the transmission and processing capabilities required for massive data. Silicon photonics, leveraging its high integration density, low cost, and CMOS compatibility, has emerged as a powerful solution to address these challenges [1]. Silicon photonics technology enables large-scale and high-volume integrated photonic devices such as optical transceivers [2], optical I/O [3], programmable photonic processors [4], etc. The 3 dB power splitters are basic and crucial components in silicon photonics. They are widely utilized as building blocks of modulators [5,6], multiplexers [7], optical switches [8], on-chip spectrometers [9], and optical phased arrays (OPAs) [10]. The traditional 3 dB power splitters generally work in a specific wavelength band. However, for broadband applications such as wavelength division multiplexing (WDM) and spectrometers, the overall performance is highly dependent on the operation bandwidth of the utilized power splitters. Additionally, in high-density applications such as OPAs and optical computing networks [11], numerous power splitters are cascaded or interconnected, which places stringent requirements on footprint and insertion loss (IL) of the splitters. Therefore, an optimal 3 dB power splitter should be broadband, compact, low-loss, and fabrication friendly. Various 3 dB power splitters have

been proposed and have partially achieved these goals. Among these power splitters, directional couplers (DCs) feature low loss and arbitrary power splitting ratios [12,13]. However, DC-based splitters are highly sensitive to wavelength and coupling length variations. Bent DCs can operate at wider wavelength ranges, but precise control of the waveguides and gap widths are required [14,15]. Y-branches are utilized as compact and polarization insensitive power splitters. However, the small branching angle and shape corners of Y-branches are challenging to fabricate [16,17]. Power splitters based on multimode interferometers enable an arbitrary number of inputs and outputs; however, they generally suffer from large device sizes and limited optical bandwidths [18,19]. Additionally, several subwavelength grating (SWG) structures have been combined with DCs and Y-branches, demonstrating compact footprints and broadband power splitting capabilities [20,21]. However, SWG structures typically require feature sizes below 100 nm, and this exceeds the resolution limits of commercial fabrication processes, potentially compromising their practical performance.

As another frequently used 3 dB power splitter, the adiabatic power splitter is inherently low loss and broadband [22,23]. This splitter typically demands sufficiently extended coupling regions for effective mode transformation [24,25]. Several schemes have been reported to realize 2×2 3 dB adiabatic power splitters with shorter lengths, such as shortest mode transformer method [26], fast quasi-adiabatic approach [27,28], and adiabaticity engineering techniques [29]. Additionally, 1×2 3 dB adiabatic power splitters possess symmetric and simple structures, enabling both compact device dimensions and broadband operation. In [30], an ultra-compact 1×2 adiabatic power splitter was reported, which has an adiabatic taper length of only 5 μm and measured in the wavelength range from 1530 to 1600 nm. However, the minimum width and gap of the tapers in the splitter were 30 nm and 50 nm, beyond common fabrication constraints. In [31], an ultra-broadband 3 dB adiabatic power splitter based on modal engineering was demonstrated from 1260 to 1650 nm, while the total length of this splitter was about 200 μm . In [32], a 3 dB adiabatic power splitter based on nonlinear taper profile was designed that had a minimum feature size of 120 nm and a length of 14 μm . Although this splitter balanced manufacturability and small size, its optical bandwidth was only 100 nm. Novel power splitter designs are imperative to enhance their operational bandwidth for broadband system integrations while minimizing device footprint and IL for the scalable deployment in high-density photonic applications.

In this work, we propose and demonstrate a compact and ultra-broadband 1×2 3 dB power splitter on a commercial 220 nm silicon-on-insulator (SOI) platform. The power splitter is based on three segmented adiabatic tapered rib waveguides (WGs), with a coupling length of 23.4 μm . The simulations show that the power splitter maintains an output transmission efficiency higher than 48% throughout the entire 1200 to 1600 nm wavelength range. In the wavelength ranges of 1260 to 1360 nm and 1525 to 1600 nm, the measured power splitting ratios of the device are both within $50 \pm 3.8\%$, and the ILs are both below -0.38 dB. Compared to conventional adiabatic power splitters, this splitter fully leverages the broadband advantages while achieving effective optimization in device footprint and IL owing to the refined structural design. Additionally, a high-speed Mach–Zehnder modulator (MZM) is constructed based on this 3 dB power splitter. The modulator demonstrates 40 Gbps NRZ signal modulation across both O-band and C-band wavelengths, exhibiting remarkable spectral compatibility. The broadband splitter and MZM are promising developments for building 400 Gbps and 800 Gbps silicon photonics integrated transceivers with WDM designs [33,34].

2. Design and Simulation

Most reported 1×2 3 dB power splitters employ linearly tapered WG structures [23,30,35]. To extend the optical bandwidth of the splitters, particularly toward the shorter wavelengths, the linear taper lengths need to be increased to ensure adiabatic mode evolution and conversion between the tapers. Similarly, for the edge coupler (EC), another fundamental component in silicon photonics, the optimization of the tapers in ECs serves as a critical approach to improve the coupling efficiency, expand the operational bandwidth, and reduce the device footprint. Various taper-based ECs have been proposed, including linear tapers [36], multi-layer tapers [37], bi-level tapers [38], segmented tapers [39,40], and SWG tapers [41,42]. The EC based on segmented tapers is exemplified in Figure 1a. This EC employs a single-layer taper structure, eliminating the need for additional material etching or growth processes required in bi-level tapers and multi-layer tapers structures. This EC also avoids the fabrication challenges of ultranarrow tip features inherent to SWG-based approaches. Compared to the EC based on one simple linear taper, through optimizing the taper length and angle of each segment, this EC maintains low-loss and broadband performance while achieving significant length reduction. Inspired by this approach, we implement a 1×2 3 dB power splitter based on segmented adiabatic tapers, as shown in Figure 1b,c. The power splitter consists of an input WG, three segmented taper sections, and two 90° output bends. Each taper section comprises three sequentially connected tapers with decreasing widths (w_0 , w_1 , w_2 , and w_3). The corresponding segmented taper lengths are marked as L_1 , L_2 , and L_3 . The width w_0 is selected as 400 nm for single-mode WG, and w_1 , w_2 , and w_3 are set as 200, 180, and 150 nm, respectively, for efficient taper transitions. The minimum gap between these segmented tapers is selected as 250 nm. All these taper widths and gaps are within the manufacturing capabilities of common silicon photonics platforms.

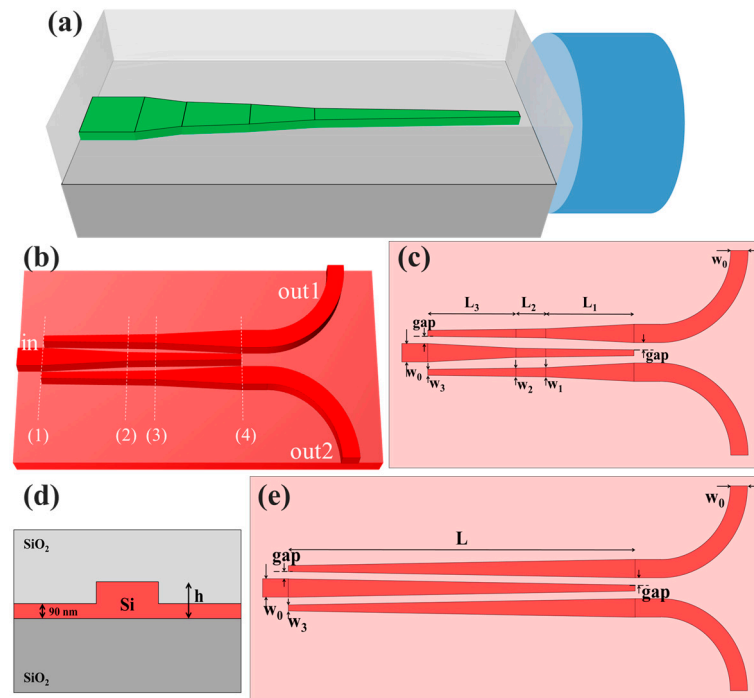


Figure 1. (a) Schematic of the edge coupler with segmented tapers. (b) The 1×2 3 dB power splitter based on segmented adiabatic tapers. (c) The parameters of the proposed power splitter. (d) The cross section of the rib waveguides. (e) A 1×2 3 dB power splitter based on linear tapers.

Moreover, the 1×2 3 dB power splitters are based on rib WGs with slab thickness of 90 nm, as shown in Figure 1d. Compared to the power splitters based on strip WGs, the rib structures mitigate the optical confinement between tapered WGs, enhancing mode coupling, and consequently enables decreasing device lengths [23]. Additionally, the power splitters based on rib WGs are more suitable for building the MZM, in which doped rib WGs typically serve as phase shifting arms [43]. For the single segmented taper in the edge coupler, as shown in Figure 1a, a narrower taper segment typically requires a longer minimum adiabatic length under the given width selections. For the designed power splitter composed of three taper segments, considering the symmetry of the power splitting process and the splitter structure, we can slightly enlarge the widest taper segment, i.e., taking $L_1 = L_3 = L_{13}$. The power splitting ratio (SR) of the 1×2 3 dB splitter is defined as follows:

$$SR_{1,2} = \frac{P_{1,2}}{P_0} \quad (1)$$

where $SR_{1,2}$ is the SR of the two output ports (“out1” and “out2”) when the light is injected through the “in” port, as marked in Figure 1b. P_0 , P_1 , and P_3 are the monitored powers at the three ports. Using the eigenmode expansion solver from Ansys Lumerical 2020 R2 software, the $SR_{1,2}$ of the power splitter are calculated at different L_{13} and L_2 . Figure 2 presents the simulated $SR_{1,2}$ at 1310 nm and 1550 nm wavelengths, respectively. Owing to the structural symmetry of the 1×2 splitter, both output ports exhibit identical SR. Therefore, the plotted data represent the performance of either output port. The lengths of the segmented tapers of the splitter are depicted as star markers in Figure 2, realizing power splitting ratios exceeding 49.3% at both wavelengths. The corresponding lengths are 10, 3.4, and 10 μm for L_1 , L_2 , and L_3 , respectively, yielding a total taper length of 23.4 μm .

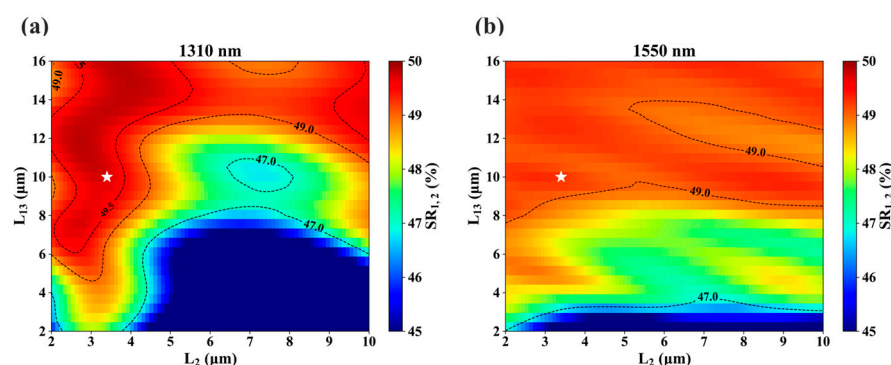


Figure 2. The power splitting ratios $SR_{1,2}$ of the 1×2 splitter with different taper lengths at (a) 1310 nm and (b) 1550 nm wavelengths, respectively.

We employed a 3D finite-difference time-domain (3D-FDTD) solver to numerically investigate the transmissions of the device. As illustrated in Figure 3a,b, the modal field distributions of the power splitter at 1310 nm and 1550 nm wavelengths present effective power splitting. When the light is injected from the left input port, it is equally divided into two beams that emerge from the upper and lower output ports, respectively. Figure 3c–j illustrates the modal field distributions at four cross-sections when the middle taper widths are w_0 , w_1 , w_2 , and w_3 , corresponding to positions (1)–(4), respectively, in Figure 1b. At both 1310 nm and 1550 nm wavelengths, the taper width in the middle of the splitter decreases while taper widths on both sides increase along the propagation direction. This geometric modulation progressively weakens the modal confinement in the central region of the splitter, initially expanding the modal field into both sides. Consequently, the modal field evolves into a super-mode distribution among three coupled WGs and gradually transforms into confined modal fields of the two output WGs.

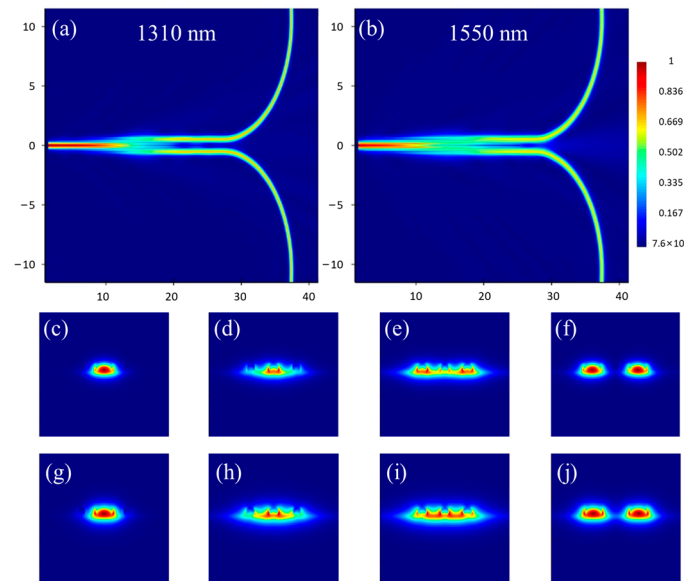


Figure 3. The modal field distributions of the power splitter at (a) 1310 nm and (b) 1550 nm wavelengths. The modal field distributions along the splitter at (c–f) 1310 nm and (g–j) 1550 nm wavelengths, corresponding to positions (1)–(4) in Figure 1b.

The IL of the 1×2 3 dB splitter is defined as follows:

$$IL = 10 \log \left(\frac{P_1 + P_2}{P_0} \right) \quad (2)$$

The SRs and ILs of the 1×2 power splitter are then calculated using the 3D-FDTD solver throughout a wide wavelength range of 1200 to 1600 nm, as shown in Figure 4a,b. Consistent with the simulations above, the spectra of the two outputs of the designed splitter are identical due to the symmetric 1×2 splitting structure. Over the 400 nm wavelength range, both output ports demonstrate SRs exceeding 48% and ILs below -0.2 dB.

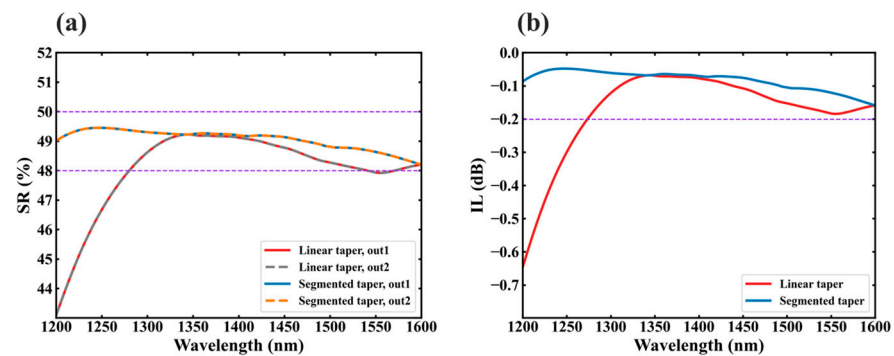


Figure 4. The simulated (a) SRs and (b) ILs of the proposed 1×2 power splitter across the wavelength range from 1200 to 1600 nm, compared to a power splitter based on linear tapers.

In Figure 4, a comparative analysis is also conducted with a power splitter based on linear tapers. As shown in Figure 1e, the compared splitter possesses a $40 \mu\text{m}$ taper length, 150 nm tip widths, and 250 nm gaps, respectively. Within the 1370 to 1420 nm wavelength range, the two splitters achieve comparable performance. Beyond this spectral window, the proposed splitter exhibits superior SR and IL. Notably, the linear taper-based splitter suffers severe degradation below the 1260 nm wavelength, demonstrating a critical limitation in short-wavelength operation. The SR spectra of the designed power splitter

affected by variations in dimensions are presented in Figure 5. The splitter demonstrates robust fabrication tolerance, maintaining SR above 47% across the 400 nm wavelength range even with ± 20 nm deviations in critical structural parameters, including waveguide widths, thicknesses, and gaps.

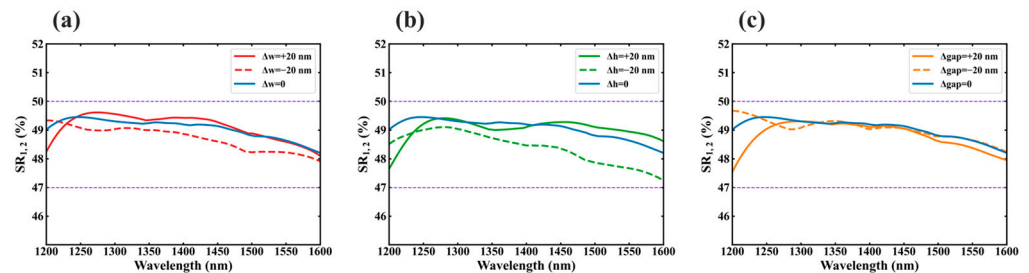


Figure 5. The SR spectra of the designed power splitter affected by (a) widths, (b) thicknesses, and (c) gaps variations. The plots show simulated results of either output of the splitter due to the identical SRs for two outputs.

Furthermore, an MZM based on the compact and broadband power splitter is constructed, as indicated in Figure 6a. The electro-optical phase shifting arms of the MZM are composed of doped rib WGs and traveling wave electrodes, as shown in Figure 6b. The width, height, and slab thickness of each rib WG are 400 nm, 220 nm, and 90 nm, respectively, same as the WG dimensions in the proposed splitter. The concentrations of the p-doped and n-doped regions (P and N) are $1 \times 10^{18}/\text{cm}^3$ and $8 \times 10^{17}/\text{cm}^3$, respectively. Both heavy p-doping and n-doping (P++ and N++) concentrations are $5.5 \times 10^{20}/\text{cm}^3$ for the ohmic contacts. The terminators are set as 30Ω to match the characteristic impedance of the traveling wave electrodes. The resistivity of the silicon substrate is $750 \Omega \cdot \text{cm}$ and the thickness of the cladding oxide is about $3 \mu\text{m}$. A push-pull configuration is selected for the MZM. This configuration provides the advantages in enhancing the linearity [44], improving the modulation efficiency [45], decreasing the inter-device crosstalk [46], and facilitating the integration with the common drivers [47].

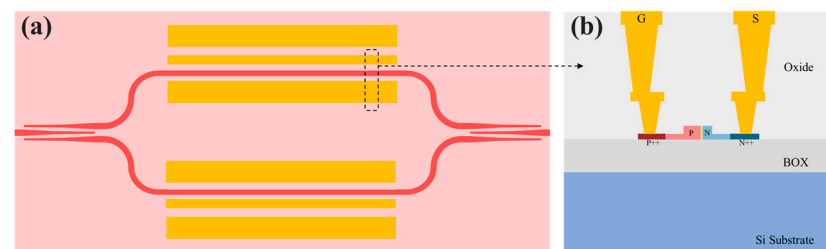


Figure 6. (a) The MZM based on the proposed 1×2 3 dB power splitter. (b) The electro-optical phase shifting arm of the MZM.

3. Fabrication and Characterization

Using a 193 nm dry lithography process, the devices are fabricated on a 220 nm-thick SOI wafer with a $3 \mu\text{m}$ -thick buried dioxide (BOX) layer at Advanced Micro Foundry, Singapore. Figure 7a,b are the microscope images of the MZM and the 1×2 3 dB power splitter. The experimental characterization of these devices is limited to the O-band and C-band due to our measurement setup. Two amplified spontaneous emission sources (ASE, from HOYATEK, Shenzhen, China) operated in these bands, and a polarization controller (PC) and an optical spectrum analyzer (OSA, YOKOGAWA AQ6370D, Tokyo, Japan) are used to measure the IL and SR of the fabricated splitter, as shown in the blue region of the Figure 8. The IL of the splitter is characterized by the structures depicted in Figure 7d,

which enable us to improve the measurement accuracy for devices with losses below -0.3 dB [16,48]. The IL test structures consist of two loops with different numbers of the splitters. No splitters are included in the reference loop, while 44 designed splitters are cascaded in the other loop. The spectra of these loops are shown in Figure 9a,b. The different spectra shapes come from the ASEs and grating couplers utilized in the two bands. The IL per splitter can be derived from the measured spectra and the cascaded splitter numbers. In each measurement of these loops, the wavelength resolution of OSA is set to be 20 pm and the polarization is adjusted into the transverse electric (TE) polarization by the PC. As illustrated in Figure 9c,d, the IL of the splitter is within the range of -0.4 to -0.25 dB across both the O-band and C-band. The measured ILs are slightly higher than the simulated ones, which may result from the extra scattering losses caused by sidewall roughness of small gaps between the tapers in the splitter.

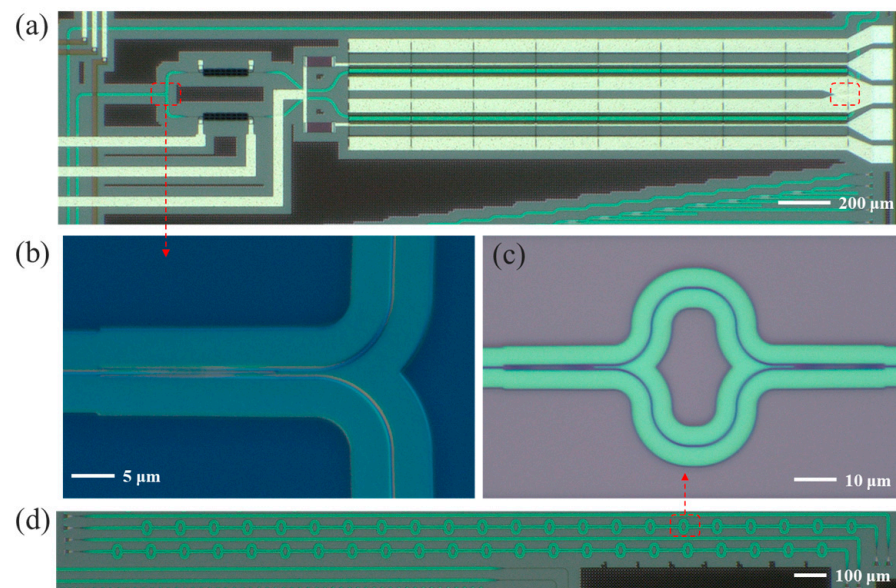


Figure 7. Microscope images of the fabricated devices. (a) The designed MZM. (b) The 1×2.3 dB power splitter. (c) A cell of the IL test structures. (d) The IL test structures of the splitter.

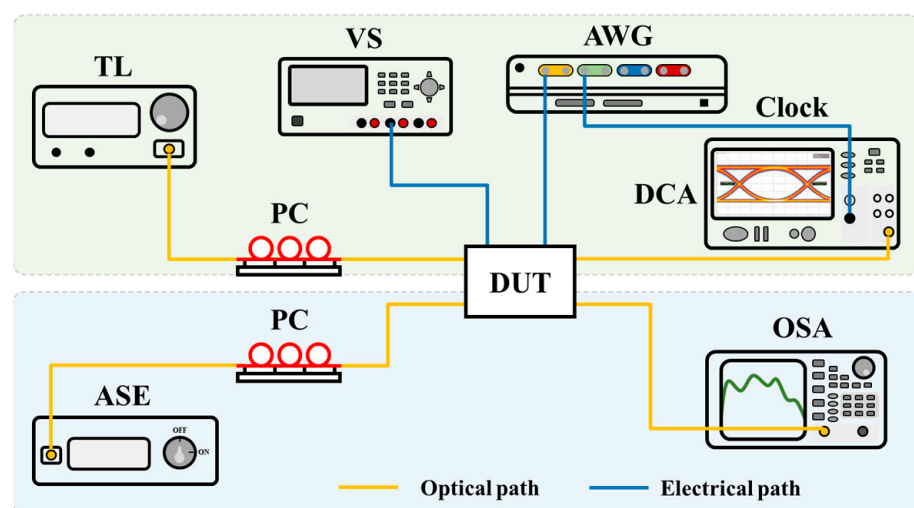


Figure 8. Experimental setup for characterizing the devices. ASE, amplified spontaneous emission; PC, polarization controller; DUT, device under test; OSA, optical spectrum analyzer; TL, tunable laser; AWG, arbitrary waveform generator; VS, voltage source; DCA, digital communication analyzer.

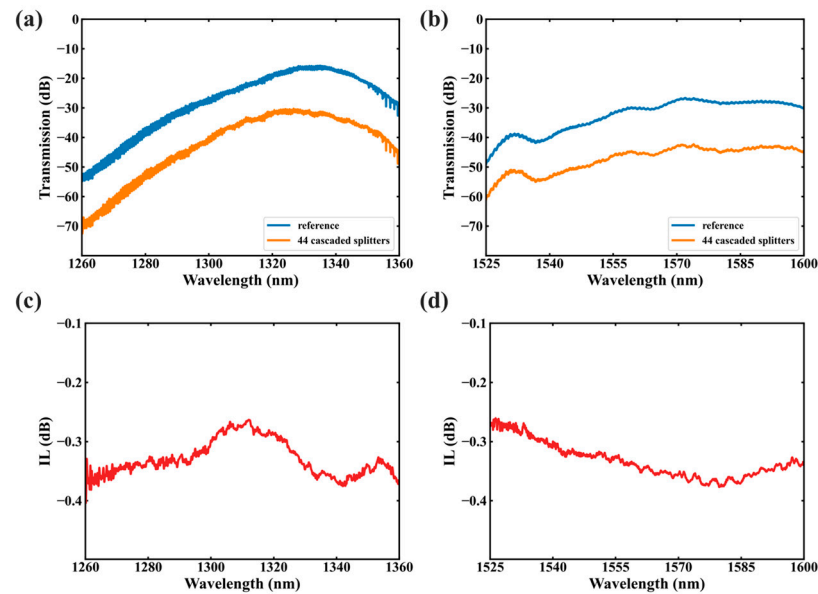


Figure 9. Measured spectra of the IL test loops in the (a) O-band and (b) C-band. The derived IL per splitter in the (c) O-band and (d) C-band.

A Mach–Zehnder interferometer (MZI) with an arm length difference of 350 μm is fabricated to obtain the SR of the designed 3 dB power splitter. For an imbalanced MZI, assuming that the 3 dB power splitters and arm WGs are lossless, the extinction ratio (ER) of the spectra relates to the SR of the splitters through the following equation [32]:

$$\text{SR}_{1,2} = \frac{1}{2} \pm \frac{1}{2} \sqrt{10^{-\text{ER}(\text{dB})/10}} \quad (3)$$

Although the losses of the power splitters are ignored here, the calculated SR values remain accurate [21,31], except for the additional losses in the transmission spectra of MZI compared to the ideal case. These loss values have also been used in several prior studies to evaluate the IL of the power splitter [24,32,49]. However, the losses inherently combine both MZI loss and fiber-to-chip coupling loss. Due to the low loss of our splitter, the losses are easily affected by the fiber-to-chip coupling process. Therefore, IL is accurately measured by characterizing the response of cascaded splitters as mentioned above. Figure 10a,b present the normalized transmission spectra of the MZI across the O-band and C-band. The transmission spectra are normalized with respect to those of the grating couplers and reference waveguides. The free spectral range (FSR) of the imbalanced MZI can be calculated as follows:

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L} \quad (4)$$

Compared to the O-band, the C-band exhibits longer wavelengths (λ) and lower group refractive indices (n_g), resulting in a larger FSR. The extracted ERs of the MZIs are plotted in Figure 10c,d, and the SRs of the splitter are derived by using Equation (3). The MZIs demonstrate ERs exceeding 23 dB across the entire O-band and C-band. The corresponding SRs of the proposed splitters are in the range of $50 \pm 3.8\%$. In fact, due to the presence of the arm length difference (350 μm), the losses of the phase shifting arms in the imbalanced MZI are unequal. This could lead to a slight underestimation of the ER compared to the result from a balanced MZI, as described in [22]. Overall, the designed 3 dB power splitter exhibits balanced behavior over a broad wavelength span.

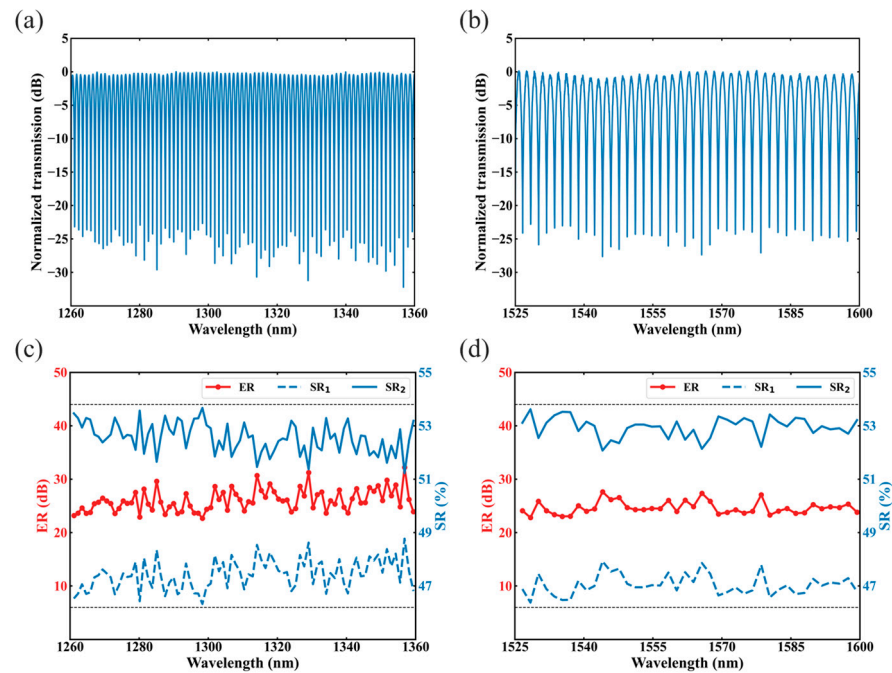


Figure 10. Measured spectra of the MZI based on the splitter in the (a) O-band and (b) C-band. The ERs of the MZI and derived SRs of the splitter in the (c) O-band and (d) C-band.

Additionally, the data transmission performance of the fabricated MZM is characterized by using the experimental setup depicted in the green region of Figure 8. A tunable laser provides continuous light at the desired wavelength, which is adjusted into TE polarization by a PC. The polarized light is fed into a lensed fiber with a 5 μm spot diameter and coupled into the chip through an EC based on an inverse taper with 200 μm length and 140 nm tip width. The pseudo-random binary sequence electrical signals are generated by an arbitrary waveform generator (AWG, Keysight M8196A, Santa Rosa, CA, USA), amplified by a linear electrical amplifier (SHF S807C, Berlin, Germany), and then loaded onto the MZM through a 67 GHz RF probe (Formfactor SP-I67-AD-GSGSG120, Livermore, CA, USA) for electro-optical modulation. The output optical signals from the chip are coupled into another lensed fiber through an identical EC. The coupling loss between the fiber and the EC is about 3 dB/facet in both bands. The modulated signals in the output fiber are finally sent to the digital communication analyzer (Agilent DCA-X 86100D, Santa Clara, CA, USA) to record the eye diagrams. The signal synchronization between the DCA and AWG is achieved via a precision clock module (Agilent 86107A, Santa Clara, CA, USA).

Before the measurement of the eye diagrams, the through-open-short-match method is used to calibrate the test setup from RF probe tips to the DCA ports with an impedance standard substrate (Cascade, 129–239, Beaverton, OR, USA). Figure 11 shows the eye diagrams of the proposed modulator operating at 1270, 1290, 1310, 1330, 1525, 1535, 1550, and 1570 nm wavelengths under 40 Gbps none-return-to-zero (NRZ) signal modulation. In the measurements, the voltage source provides the direct current voltages for the modulator. The bias voltage of the MZM is set into -2 V as the designed MZM works in the carrier-depletion mode. The differential driving signal amplitude applied to the MZM is 5 V_{pp} . The eye diagrams at eight wavelengths are all open and clear, and the dynamic extinction ratios are measured as 6.5, 7.6, 7.9, 7.8, 7.3, 6.6, 6.7, and 6.2 dB, respectively. The eye crossings are at the position of $50 \pm 0.7\%$ with little jitters. These results validate the capability of the MZM to support 40 Gbps high-speed electro-optic modulation across both O-band and C-band.

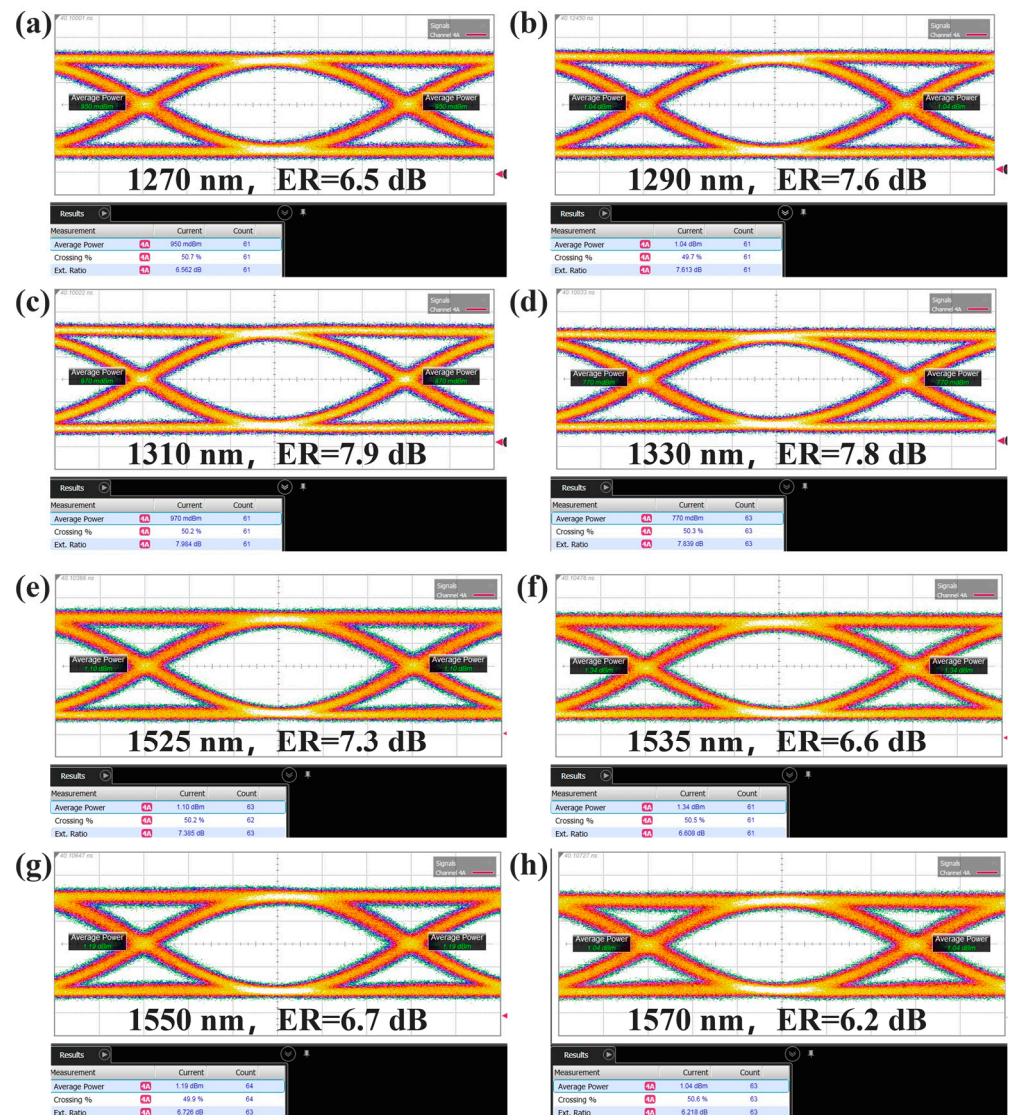


Figure 11. 40 Gbps NRZ eye diagrams for the designed MZM at the wavelength of (a) 1270, (b) 1290, (c) 1310, (d) 1330, (e) 1525, (f) 1535, (g) 1550, and (h) 1570 nm.

4. Discussions

Table 1 summarizes performances of the reported 1×2 3 dB adiabatic power splitters and this work. The proposed 3 dB adiabatic power splitter is fabricated by a lithography technology with a critical dimension (CD) of 150 nm, realizing a compact coupling length of 23.4 μm and low losses below 0.4 dB within a 175 nm optical bandwidth in the measurement. The device reported in [23] employs a similar fabrication process, but its length is 1.7 times longer while the measured optical bandwidth is only 57% of the proposed splitter. The splitters in [30,32] possess smaller footprints and ILs, and their bandwidths are below 100 nm. Moreover, these splitters are only demonstrated by using electron beam lithography with narrower taper tips and gaps, which poses significant challenges for high-volume manufacturing due to the throughput and fabrication uniformity. For the splitter reported in [31], an impressive operation bandwidth of 390 nm is demonstrated. However, the device length of 200 μm and CD of 100 nm impose limitations on its scalability potential in high-density photonic integrated circuits. Limited by the available experimental setups, our device is measured in the O-band and C-band. Due to the good agreement between the measurement and simulation, our device is also expected to demonstrate a 400 nm bandwidth as the simulated result. Inspired by the segmented structures employed in

several high-performance ECs, the power splitter is designed based on three segmented adiabatic tapers. To the best of our knowledge, this is the first implementation of such a structure in 3 dB power splitter design, which effectively achieves a compact footprint, a large bandwidth, and great manufacturability. The design methods introduced in this work can also be applied to other materials such as silicon nitride, aluminum nitride, and lithium niobate.

Additionally, several broadband optical switches based on 3 dB power splitters with large bandwidths have been proposed or demonstrated in prior works [50,51]. The simulations of the optical switch based on SWG splitters in [50] show a worst ER of 13 dB and a worst IL of 2 dB over a bandwidth of 150 nm. The optical switch in [51] is formed by MMI-based splitters. This switch experimentally demonstrated an ER higher than 15 dB and an IL less than 2.5 dB in the wavelength range of 1520 to 1580 nm. Based upon the experimental verification of the low IL (<0.4 dB) and uniform splitting performance ($50 \pm 3.8\%$) in the ranges of 1260 to 1360 nm and 1525 to 1600 nm, our proposed splitter is believed to be capable of significantly advancing the broadband optical switches. In addition to the broadband MZM shown in this work, the proposed 3 dB splitter can be extended to other applications requiring broadband operation such as WDM systems [33,34] and OPAs [52,53]. The wavelength-agnostic and low loss performance of the proposed splitter can address critical challenges in these systems, including the demand for high channel-count scalability in WDM architectures and phase consistency across OPA elements.

Table 1. Comparison of 1×2 3 dB adiabatic power splitters reported in recent years.

Ref.	Length (μm)	CD ¹ (nm)	Bandwidth (nm)	IL (dB)	Fabrication
[23]	40	200	1470–1570 ^M	<0.1	lithography
[30]	5	30	1200–1700 ^S /1530–1600 ^M	<0.19	EBL ²
[31]	200	100	1260–1650 ^M	<0.5	EBL
[32]	14	120	1500–1600 ^M	<0.25	EBL
[54]	20	180	1480–1585 ^M	<0.5	lithography
This work	23.4	150	1200–1600 ^S /1260–1360 ^M and 1525–1600 ^M	<0.4	lithography

^{S,M} These superscripts mean simulated or measured bandwidths. ¹ CD: critical dimension. ² EBL: electron beam lithography.

5. Conclusions

In summary, we proposed and experimentally demonstrated a compact and ultra-broadband 3 dB power splitter on a commercial 220 nm SOI platform. The power splitter was composed of three segmented tapered rib WGs, realizing an adiabatic coupling length of 23.4 μm . Based on the optimization of the segmented structures, the simulated power splitter exhibited $>48\%$ splitting efficiency with <0.2 dB IL in an unprecedented wavelength bandwidth of 400 nm, ranging from 1200 to 1600 nm. The fabricated device demonstrated the SRs of $50 \pm 3.8\%$ and the ILs of 0.25–0.4 dB across the O-band and C-band. Additionally, a broadband MZM based on the splitter was designed. At eight typical wavelengths of the O-band and C-band, the modulator demonstrated 40 Gbps NRZ high-speed signal modulation with dynamic ERs exceeding 6.2 dB. The proposed 1×2 3 dB splitter paves the way toward silicon photonic components with compact and ultra-broadband operation for next generation integrated photonic systems.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

DC	directional coupler
SWG	subwavelength grating
SOI	silicon-on-insulator
OPA	optical phased array
WDM	wavelength division multiplexing
EC	edge coupler
WG	waveguide
MZM	Mach-Zehnder modulator
SR	splitting ratio
FDTD	finite-difference time-domain
IL	insertion loss
BOX	buried dioxide
OSA	optical spectrum analyzer
ASE	amplified spontaneous emission
PC	polarization controller
TE	transverse electric
MZI	Mach-Zehnder interferometer
ER	extinction ratio
FSR	free spectral range
AWG	arbitrary waveform generator
DCA	digital communication analyzer
NRZ	none-return-to-zero
CD	critical dimension
EBL	electron beam lithography

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